Data Flow & System Behavior

MediCare - DFD, Sequence, Activity, State, and Class Diagrams

DEPI Full Stack .NET Track | Submission deadline: 6 November 2026

Overview

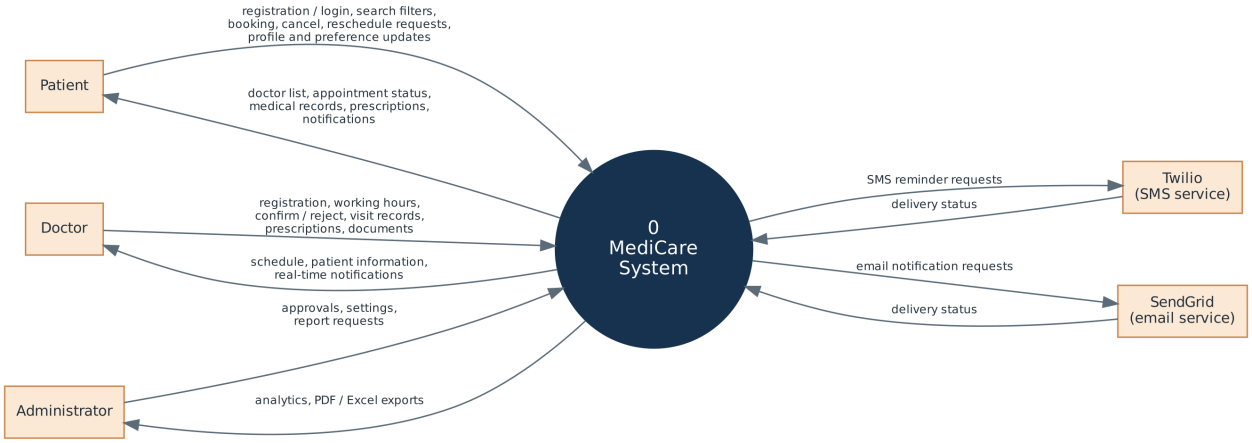
This document describes how data moves through MediCare and how the system behaves, using five kinds of diagrams: Data Flow Diagrams (DFD), Sequence Diagrams, an Activity Diagram, State Diagrams, and Class Diagrams. The diagrams follow the MediCare project proposal (roles, modules, N-Tier architecture, and technologies) and the Requirements Gathering document.

The diagrams are proposed designs. Class names, attributes, and statuses may be refined during implementation. Items that the proposal does not specify (for example the exact status list and the prescription entities) are marked as proposed in the notes at the end.

1. Data Flow Diagrams (DFD)

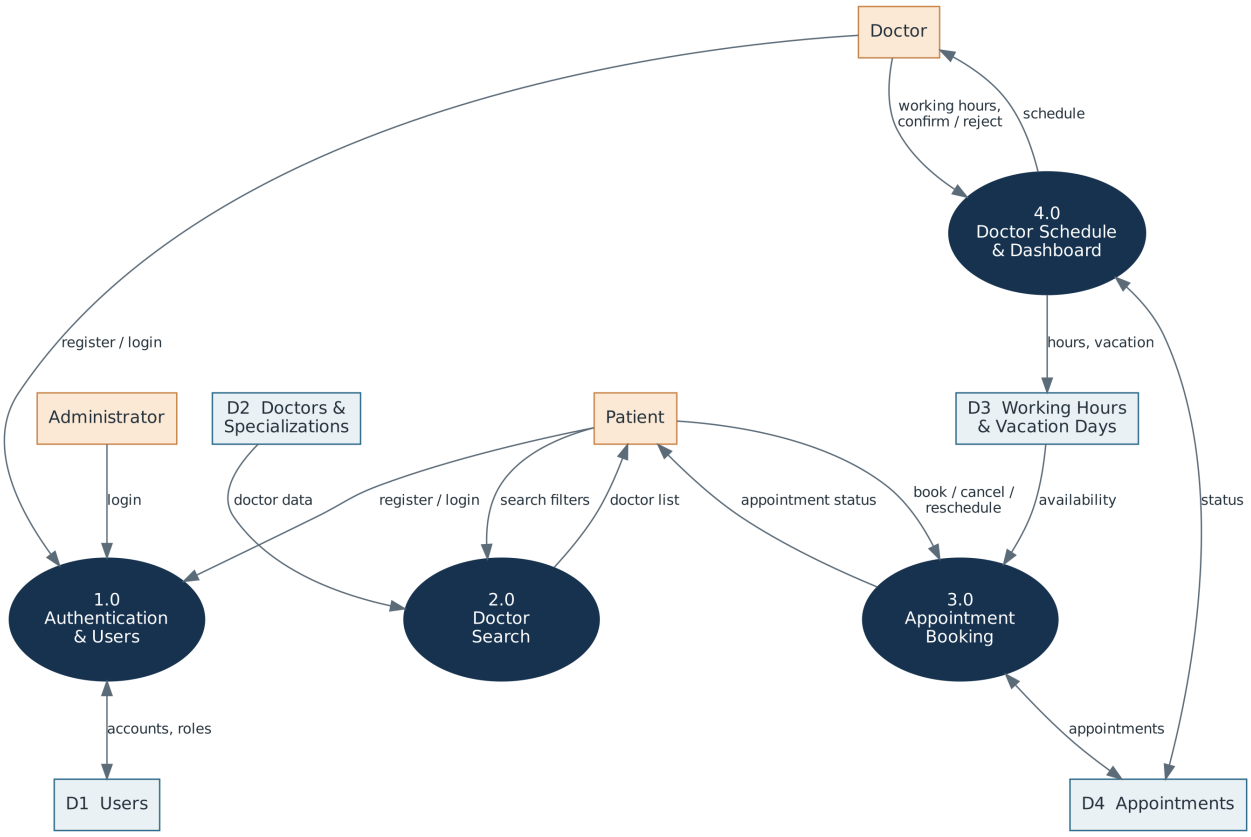
1.1 Context diagram (Level 0)

The context diagram shows MediCare as one process and the external entities that exchange data with it: patients, doctors, administrators, and the Twilio and SendGrid services.



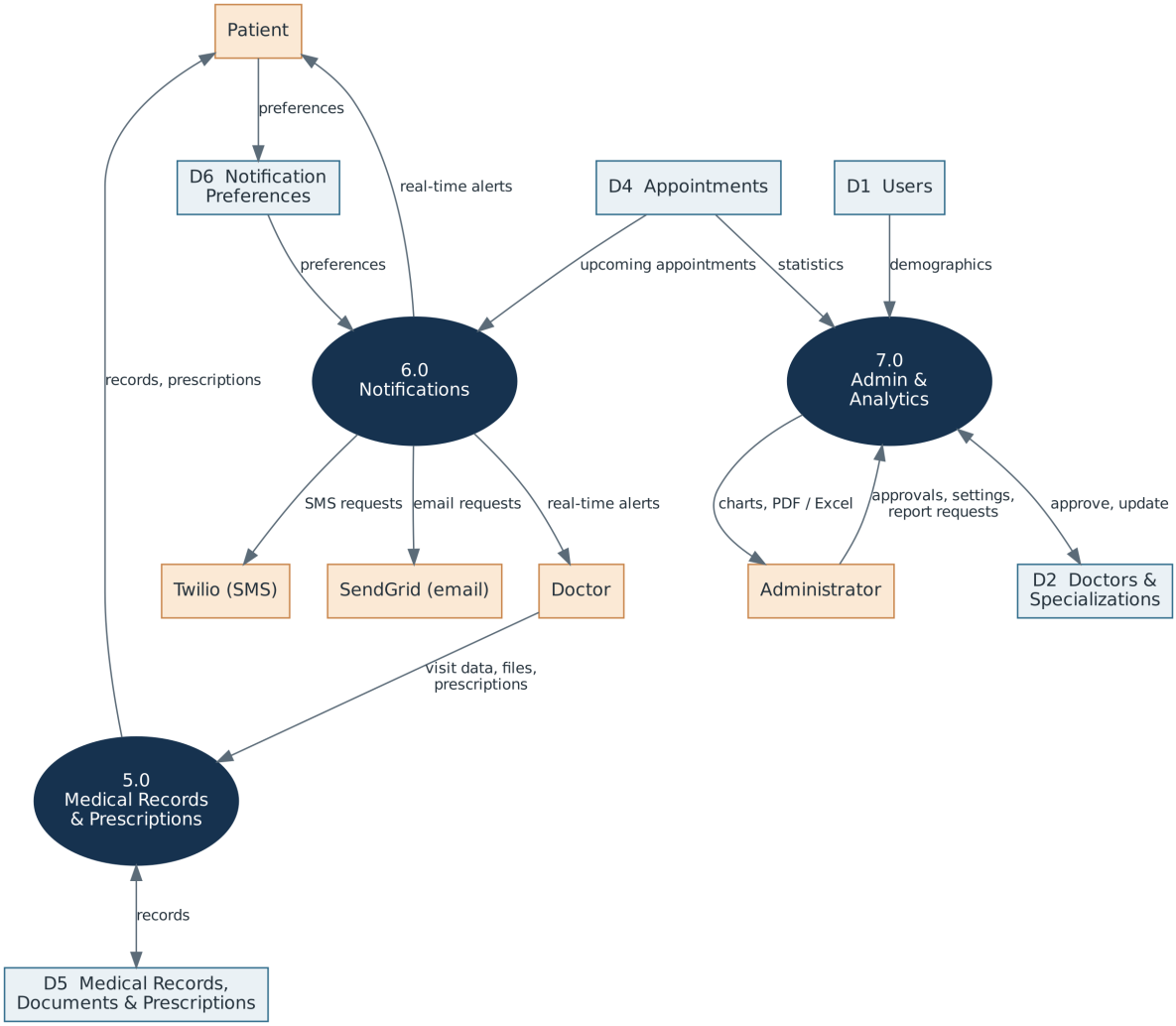
1.2 Level 1 DFD: accounts, search, booking, and schedule

Processes 1.0 to 4.0 cover authentication, doctor search, appointment booking, and the doctor's schedule and dashboard. Rectangles with a D prefix are data stores.



1.3 Level 1 DFD: records, notifications, and administration

Processes 5.0 to 7.0 cover medical records and prescriptions, notifications (real-time, SMS, and email), and administration with analytics.



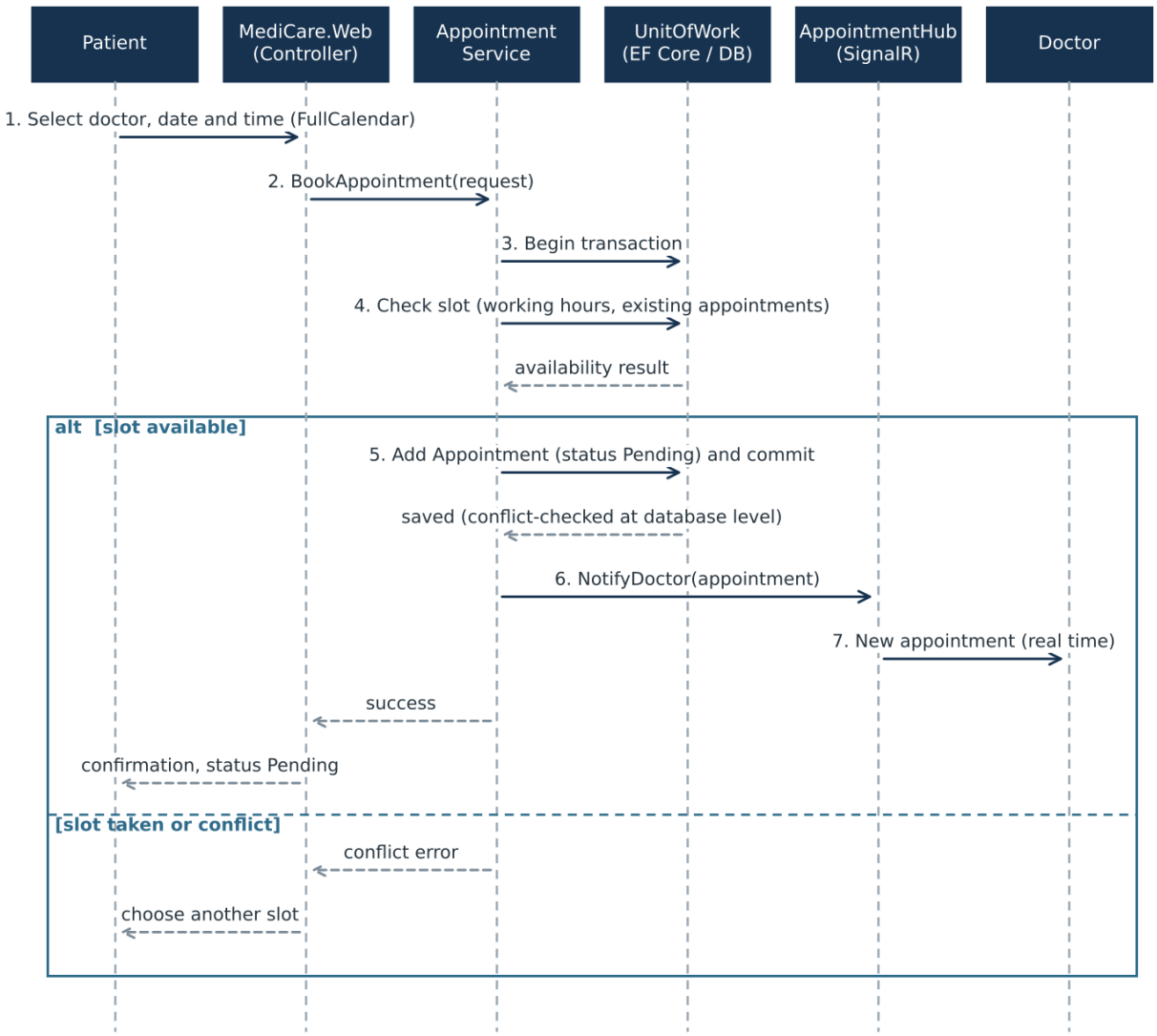
1.4 Process summary

ID	Process	Description	Data stores
1.0	Authentication & Users	Registers and authenticates patients, doctors, and administrators with ASP.NET Identity and applies role-based access.	D1
2.0	Doctor Search	Returns doctors filtered by specialization, location, and availability.	D2
3.0	Appointment Booking	Books, cancels, and reschedules appointments, using working hours for availability and checking for conflicts.	D3, D4
4.0	Doctor Schedule & Dashboard	Lets doctors manage working hours and vacation days, view schedules, and confirm or reject requests.	D3, D4
5.0	Medical Records & Prescriptions	Stores visit records, uploaded documents, and digital prescriptions, and shows them to authorized users.	D5
6.0	Notifications	Sends real-time alerts through SignalR and reminders by SMS (Twilio) and email (SendGrid), according to preferences.	D4, D6
7.0	Admin & Analytics	Approves doctors, manages settings, and produces Chart.js analytics with PDF and Excel export.	D1, D2, D4

2. Sequence Diagrams

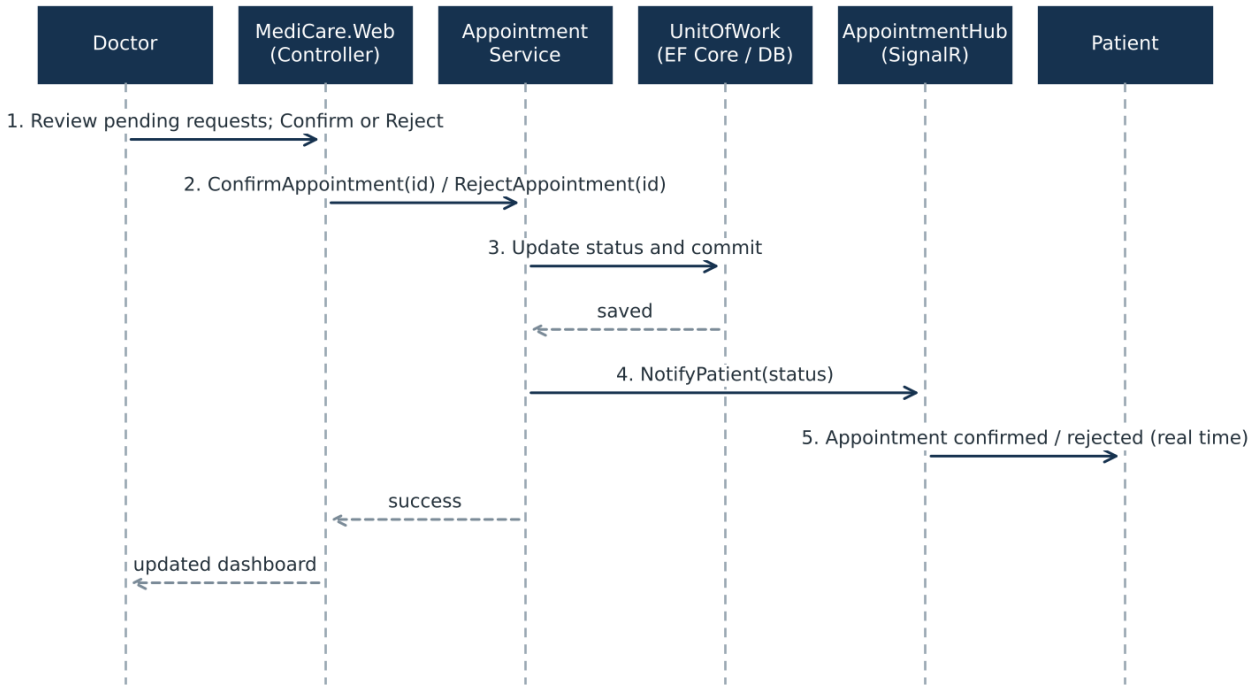
2.1 Book an appointment

The booking runs inside one transaction. The conflict check happens in the service layer and again when the data is committed, so two patients cannot hold the same slot.



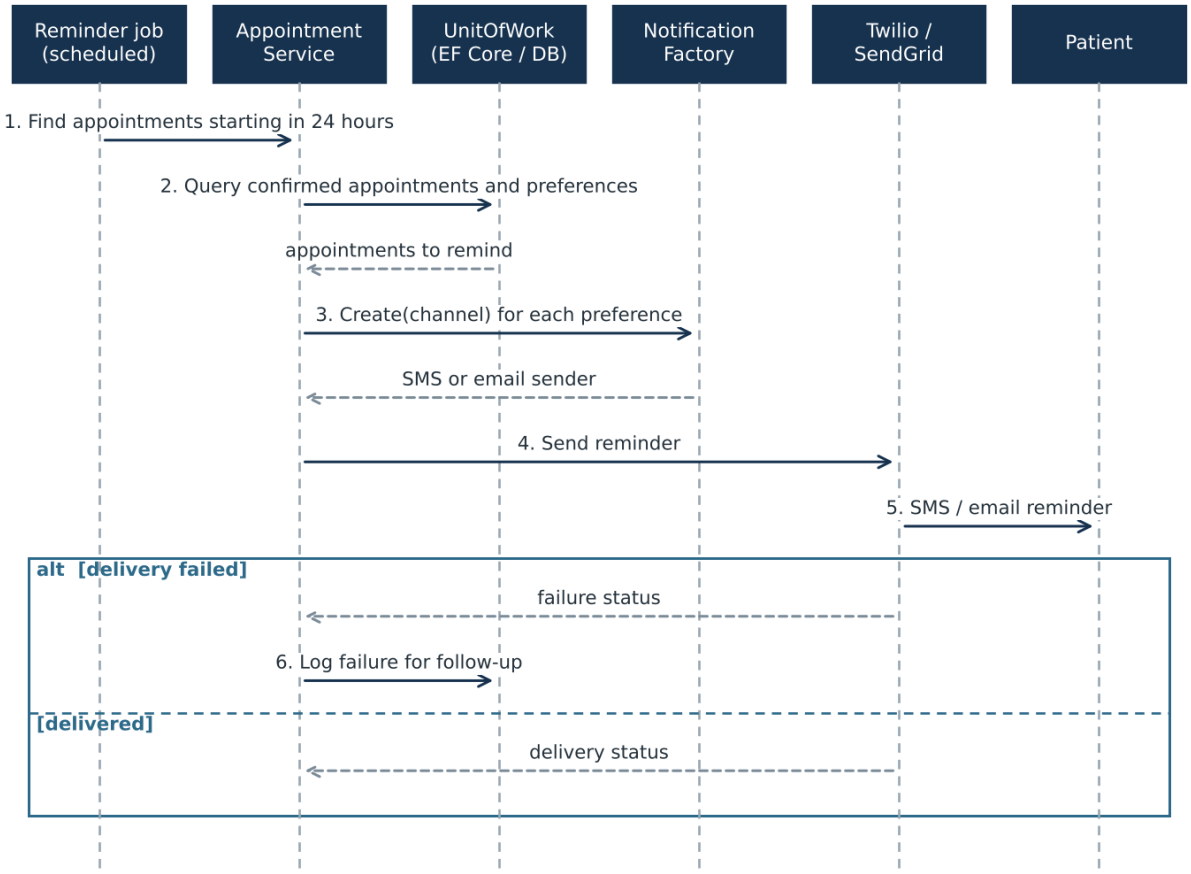
2.2 Doctor confirms or rejects a request

The doctor updates the status from the dashboard, and the patient receives the result in real time through SignalR.



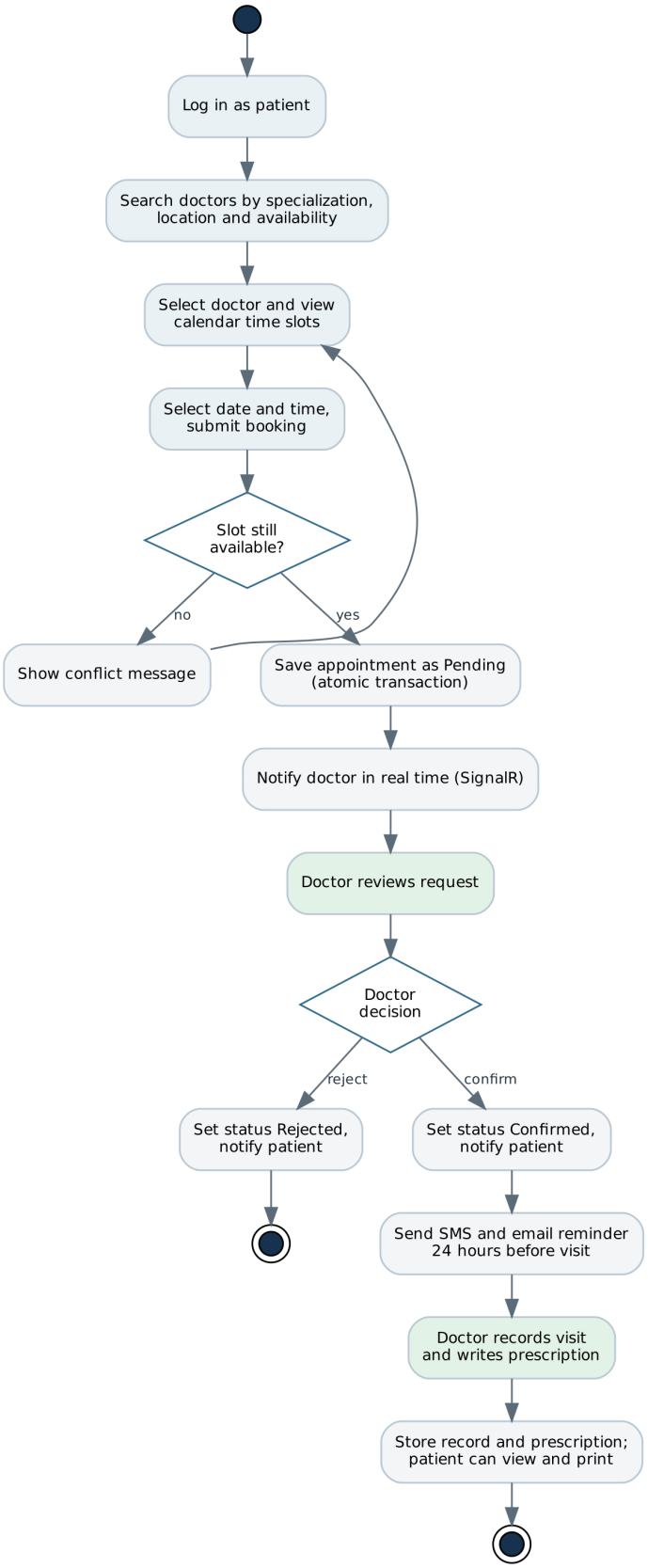
2.3 Automated 24-hour reminder

A scheduled job finds appointments starting in 24 hours, asks the notification factory for the sender that matches each patient's preference, and logs delivery failures.



3. Activity Diagram

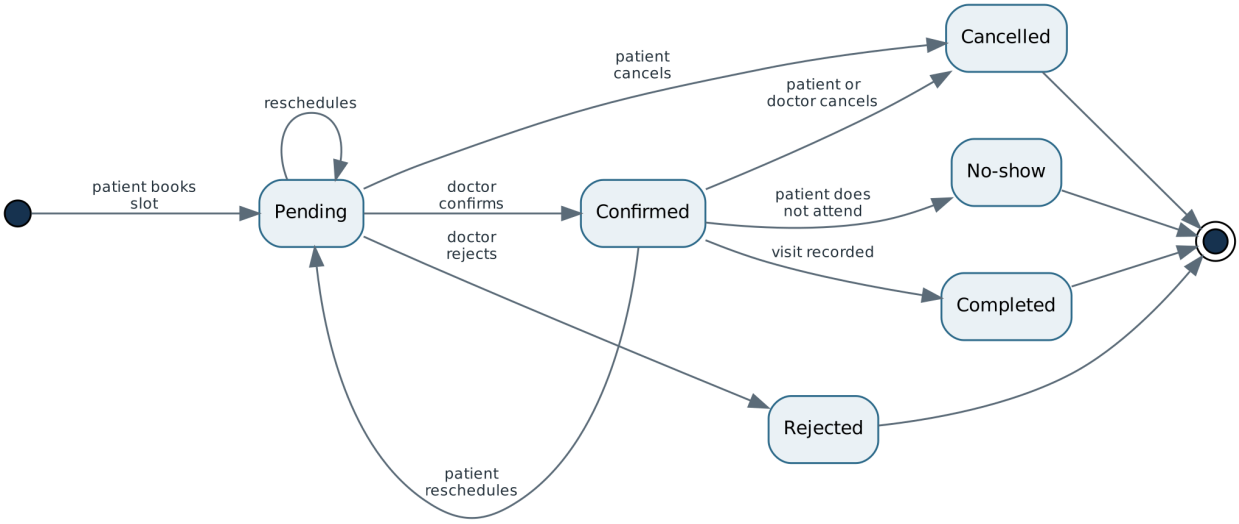
The diagram follows an appointment from the patient's login to the stored visit record. Light blue steps are performed by the patient, grey steps by the system, and green steps by the doctor.



4. State Diagrams

4.1 Appointment states

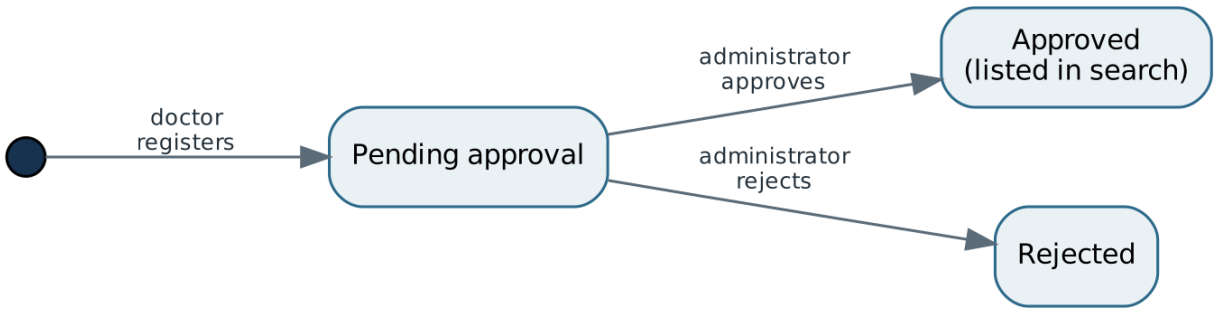
A new booking starts as Pending. The status list is a proposal and will be finalized in the design review.



From	Trigger	To	Actor
(start)	Patient books a free slot	Pending	Patient
Pending	Doctor confirms	Confirmed	Doctor
Pending	Doctor rejects	Rejected	Doctor
Pending	Patient cancels	Cancelled	Patient
Pending / Confirmed	Patient reschedules to a new free slot	Pending	Patient
Confirmed	Patient or doctor cancels	Cancelled	Patient, Doctor
Confirmed	Visit is recorded	Completed	Doctor
Confirmed	Patient does not attend	No-show	Doctor

4.2 Doctor account states

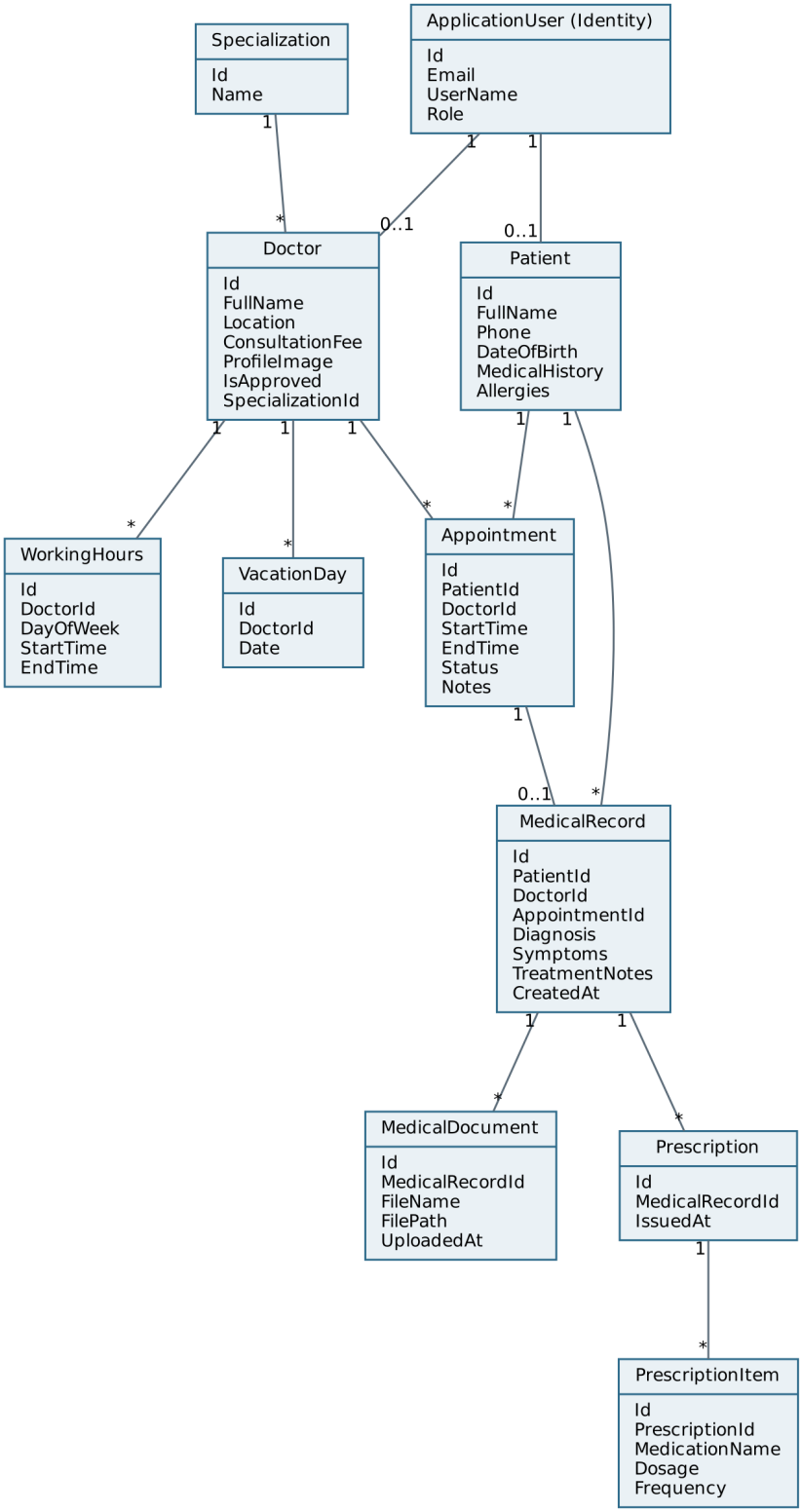
A doctor appears in search results only after an administrator approves the registration.



5. Class Diagrams

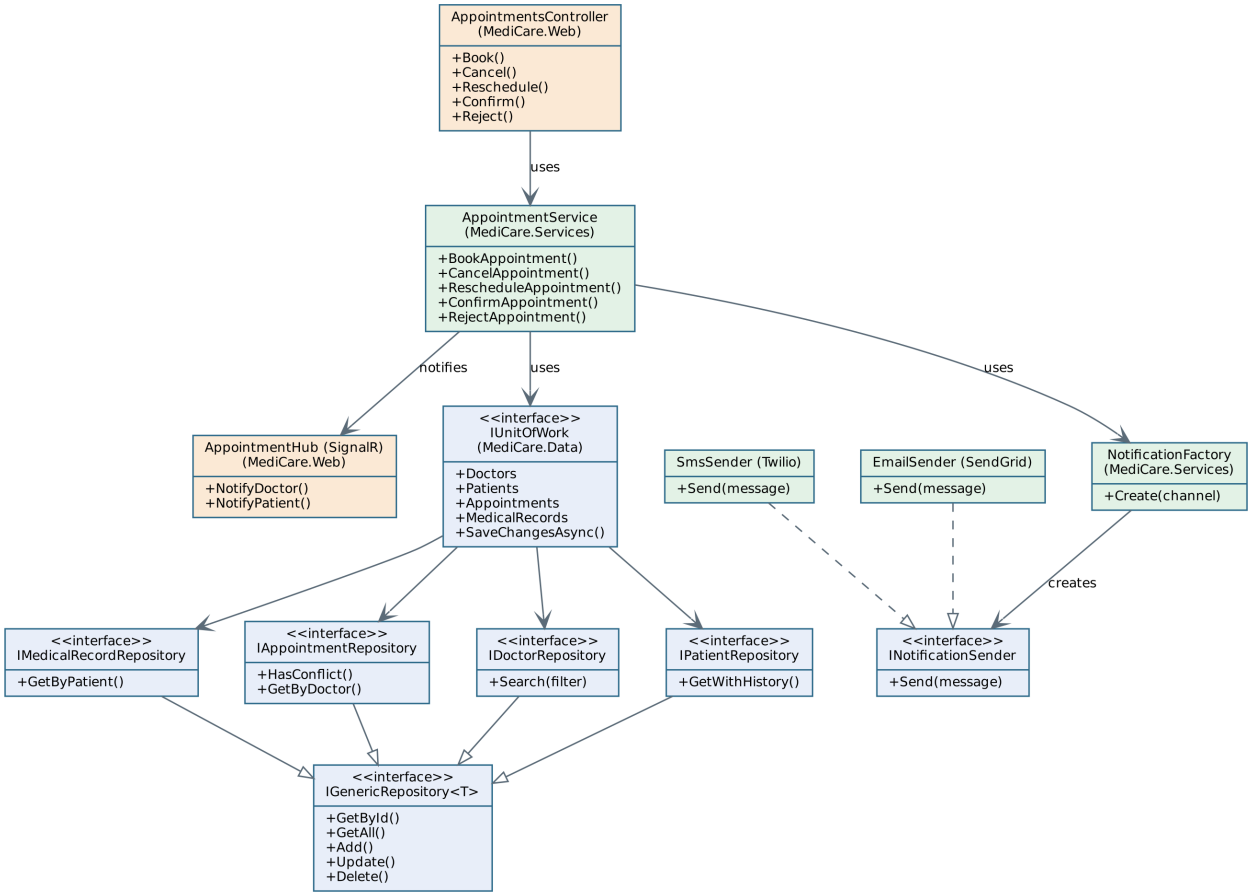
5.1 Domain model

The domain model is based on the tables listed in the proposal (Doctors, Patients, Appointments, Specializations, WorkingHours, MedicalRecords), extended with entities needed for documents, vacation days, and prescriptions.



5.2 Layers and services

The classes follow the N-Tier structure: controllers and the SignalR hub in MediCare.Web, services and the notification factory in MediCare.Services, and repositories and the Unit of Work in MediCare.Data.



Notes and Assumptions

- The diagrams are derived from the MediCare proposal and the Requirements Gathering document and have not been validated against the final implementation.
- Proposed (not specified in the proposal): the appointment status list including No-show, the Doctor account states, the VacationDay, MedicalDocument, Prescription, and PrescriptionItem entities, and the scheduled reminder job.
- Attribute lists show the main fields only. Data types, keys, and constraints will appear in the ER diagram and the logical and physical schema.